

Problem

A transistor with

$$\begin{array}{ll} h_{fe} = 120, & h_{ie} = 2 \text{ k}\Omega \\ h_{re} = 10^{-4}, & h_{oe} = 20 \mu\text{S} \end{array}$$

is used for a CE amplifier to provide an input resistance of 1.5 k Ω .

(a) Determine the necessary load resistance R_L .

(b) Calculate A_v , A_i , and Z_{out} if the amplifier is driven by a 4-mV source having an internal resistance of 600 Ω .

(c) Find the voltage across the load.

Step-by-step solution

Step 1 of 7

Consider the following data:

Base input impedance, $h_{ie} = 2 \text{ k}\Omega$

Reverse voltage feedback ratio, $h_{fe} = 120$

Base-collector current gain, $h_{re} = 10^{-4}$

Output admittance, $h_{oe} = 20 \mu\text{S}$

Input resistance, $Z_{in} = 1.5 \text{ k}\Omega$

Step 2 of 7

(a)

Write the expression for the input impedance of a common emitter transistor circuit.

$$Z_{in} = h_{ie} - \frac{h_{re} h_{fe} R_L}{1 + h_{oe} R_L}$$

Calculate the load resistance from the above expression.

$$1500 = 2000 - \frac{(10^{-4})(120) R_L}{1 + (20 \times 10^{-6}) R_L}$$

$$\frac{(10^{-4})(120) R_L}{1 + (20 \times 10^{-6}) R_L} = 2000 - 1500$$

$$(120 \times 10^{-4}) R_L = 500 [1 + (20 \times 10^{-6}) R_L]$$

$$(120 \times 10^{-4}) R_L = 500 + 0.01 R_L$$

$$0.012 R_L - 0.01 R_L = 500$$

$$0.002 R_L = 500$$

$$\begin{aligned} R_L &= \frac{500}{0.002} \\ &= 250 \text{ k}\Omega \end{aligned}$$

Step 3 of 7

(b)

Calculate the voltage gain of the transistor.

$$\begin{aligned} A_v &= \frac{-h_{fe}R_L}{h_{ie} + (h_{ie}h_{oe} - h_{re}h_{fe})R_L} \\ &= \frac{-(120)(250 \times 10^3)}{2000 + [(2000)(20 \times 10^{-6}) - (10^{-4})(120)](250 \times 10^3)} \\ &= \frac{-300 \times 10^5}{2000 + [0.04 - 0.012](250 \times 10^3)} \\ &= \frac{-300 \times 10^5}{2000 + [0.028](250 \times 10^3)} \\ &= \frac{-300 \times 10^5}{2000 + 7000} \\ &= \frac{-300 \times 10^5}{9000} \\ &= -3333 \end{aligned}$$

Step 4 of 7

Therefore, the voltage gain of the transistor, A_v is .

Step 5 of 7

Calculate the current gain of the transistor.

$$\begin{aligned} A_i &= \frac{h_{fe}}{1 + h_{oe}R_L} \\ &= \frac{120}{1 + (20 \times 10^{-6})(250 \times 10^3)} \\ &= \frac{120}{1 + 5} \\ &= \frac{120}{6} \\ &= 20 \end{aligned}$$

Step 6 of 7

Therefore, the current gain of the transistor, A_i is .

Step 7 of 7

Calculate the output impedance.

$$\begin{aligned}
 Z_{\text{out}} &= \frac{R_s + h_{ie}}{(R_s + h_{ie})h_{oe} - h_{re}h_{fe}} \\
 &= \frac{600 + 2000}{(600 + 2000)(20 \times 10^{-6}) - (10^{-4})(120)} \\
 &= \frac{2600}{0.052 - 0.012} \\
 &= \frac{2600}{0.04} \\
 &= 65 \text{ k}\Omega
 \end{aligned}$$

Therefore, the output impedance, Z_{out} is $65 \text{ k}\Omega$.

(c)

Write the expression for the voltage gain.

$$\begin{aligned}
 A_v &= \frac{V_A}{V_B} \\
 &= \frac{V_L}{V_s} \quad (\text{since } V_B = V_s \text{ and } V_A = V_L)
 \end{aligned}$$

Calculate the voltage across the load from the above expression.

$$\begin{aligned}
 V_L &= A_v V_s \\
 &= (-3333)(4 \times 10^{-3}) \\
 &= -13.33 \text{ V}
 \end{aligned}$$

Therefore, the voltage across the load, V_L is -13.33 V .